

PHYSICS-1

(15B11PH111)

Introduction

Teaching Process

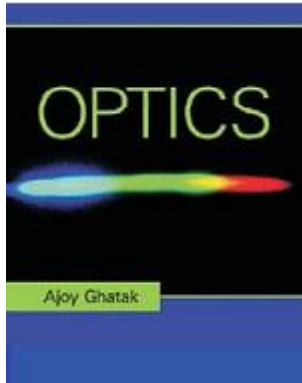
The whole course will be covered through approximately

- ☐ Lectures (40)
- ☐ Tutorials (11)

Evaluation scheme

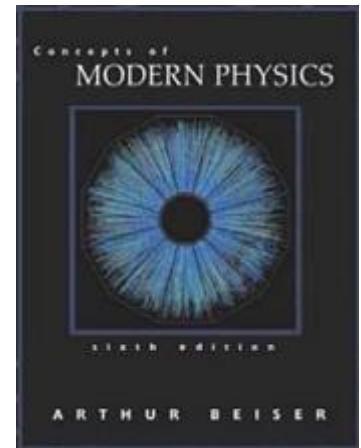
<u>Instrument</u>	<u>Marks</u>	<u>Duration</u>	<u>Description</u>
Test-1 (T1)	20	1 hr	Course covered up to T1
Test-2 (T2)	20	1 hr	Course covered between T1 and T2
END Semester Exam	35	2:00 hrs	Full course
Teacher's Assessment (TA)	25	Entire Semester	
Total	100		

Teacher's Assessment (25): PBL (10), Class Test (5), Assignments, Internal assessment, Attendance, and regularity in classes



OPTICS by [Ajoy Ghatak](#)

Concepts of Modern Physics by [Arthur Beiser](#)



Reference Books:

- E. Hecht, Optics, Pearson Education.
- F. A. Jenkins and H. E. White, Fundamentals of Optics, Tata McGraw Hill.
- A. Beiser, Concepts of Modern Physics, Mc Graw Hill International.
- Zemansky, Thermodynamics

Course Outline

1. Interference

2. Diffraction

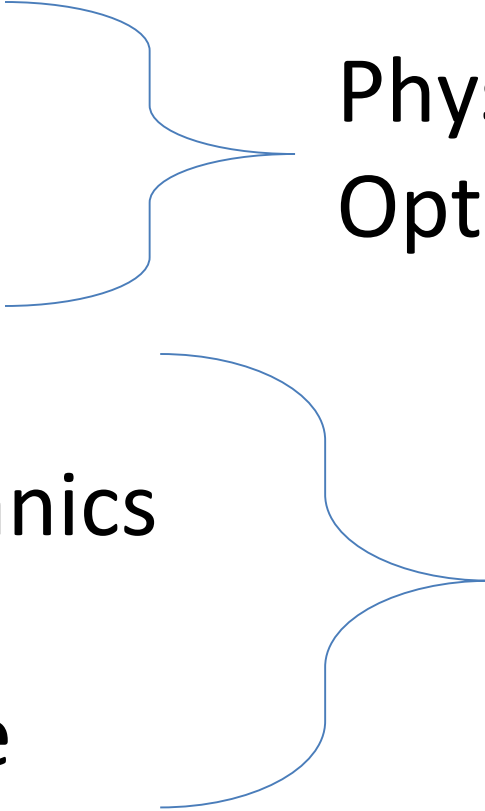
3. Polarization

4. Relativity

5. Quantum Mechanics

6. Radiation

7. Atomic Structure



Physical
Optics

The diagram uses blue curly braces to group the topics. The first brace groups items 1 through 3 (Interference, Diffraction, Polarization) under the heading 'Physical Optics'. The second brace groups items 4 through 7 (Relativity, Quantum Mechanics, Radiation, Atomic Structure) under the heading 'Modern Physics'.

Modern
Physics

Objective

- Course is designed to cover a broad aspect of physics, required for engineering student
- To improve logical thinking to solve physical problems in science & technology
- Basic knowledge for higher semester and higher education

Learning outcomes

- Sufficient understanding of wave nature of light and related phenomenon, Relativity, Quantum mechanics, Thermodynamics and their applications.

COURSE OUTCOMES:

Upon the completion of this course, students will be able to

S. No.	DESCRIPTION	COGNITIVE LEVEL
C101.1	Recall the basic principles of physics related to optics, relativity, quantum mechanics, atomic physics.	C1 (Remembering)
C101.2	Illustrate the various physical phenomena with interpretation based on the mathematical expressions involved.	C2 (Understanding)
C101.3	Apply the concepts/principles to solve the problems related to wave nature of light, relativity, quantum mechanics and atomic physics.	C3 (Applying)
C101.4	Analyze and examine the solution of the problems using physical and mathematical concepts involved.	C4 (Analyzing)

Nature of light: a wave or a particle???

The story of Light

- Corpuscular theory ([Newton](#))
- Wave nature ([Huygens](#))
- Double-slit interference experiment ([Young](#))
- Light is an electromagnetic wave ([Maxwell](#))
- Photoelectric effect –existence of light quanta – photons ([Einstein](#))
- Photons have momentum([Compton](#)).....



Wave – Particle Duality of Light

In 1924 Einstein wrote, *“There are therefore now two theories of light, both indispensable, and ... without any logical connection.”*

Evidence for wave nature of light

Diffraction, Interference, Polarization

Evidence for particle nature of light

Photoelectric effect and Compton effect

- ✓ Light exhibits diffraction and interference phenomena that are only explicable in terms of wave properties.
- ✓ Light is always detected as packets (photons); if we look, we never observe half a photon.
- ✓ Number of photons proportional to energy density (*i.e.* to square of electromagnetic field strength).

Wave

How light travel from one source reaches other location?

Waves are characterized by crest (highs) and trough (lows).

There are two types of common waves.

i) Light wave

ii) Sound wave

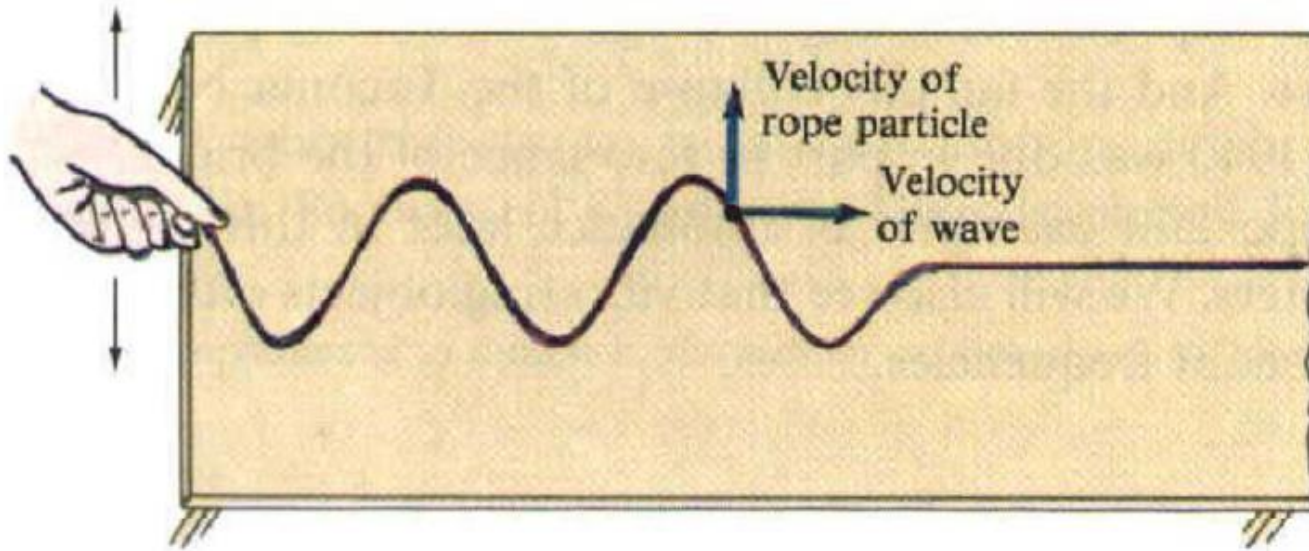
Nature of Waves

- Wave - a disturbance that travels from one point to another.
- It is a way in which energy is transmitted from place to place without physical movement of material from one place to another.

Waves are also classified as:

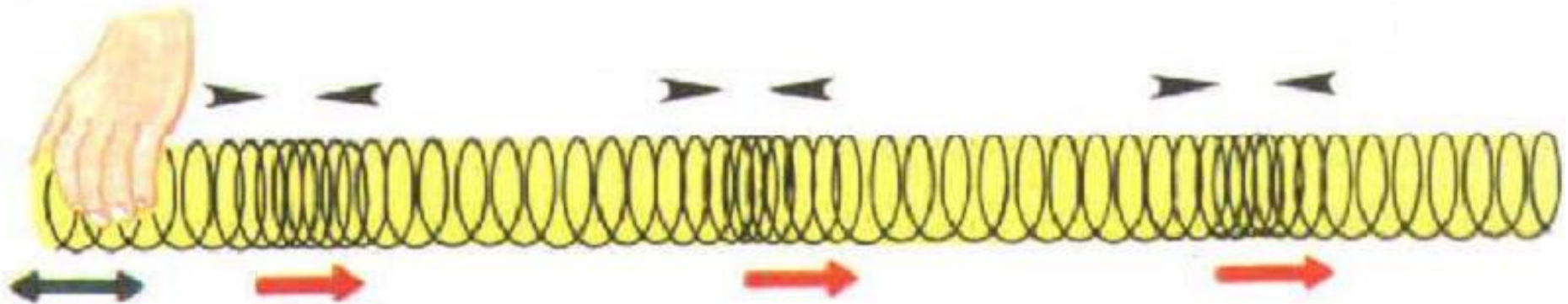
- ❖ Transverse wave
- ❖ Longitudinal wave

Transverse wave



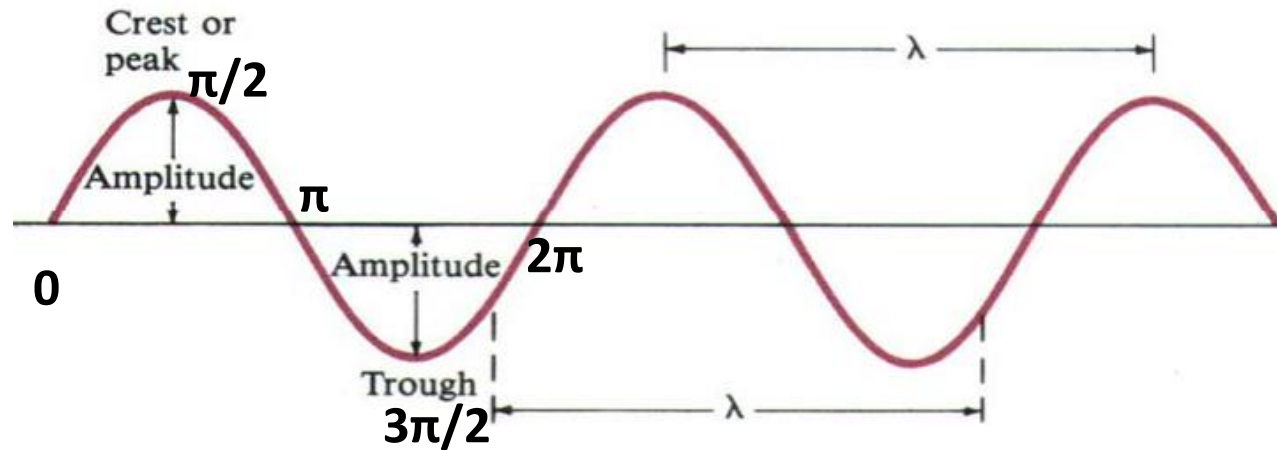
- The waveform appears in the shape of sine curve.
- A wave in which the motions of the matter particles are perpendicular to the direction of propagation of the wave itself.

Longitudinal wave

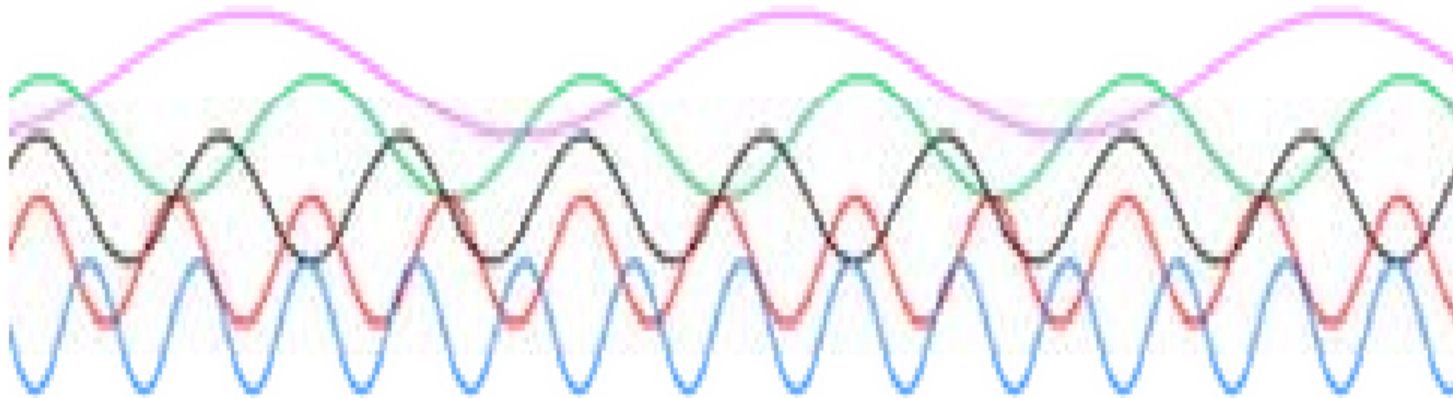


A wave in which the motions of the matter particles are in the same direction as the wave propagation.

Physical Description of a wave



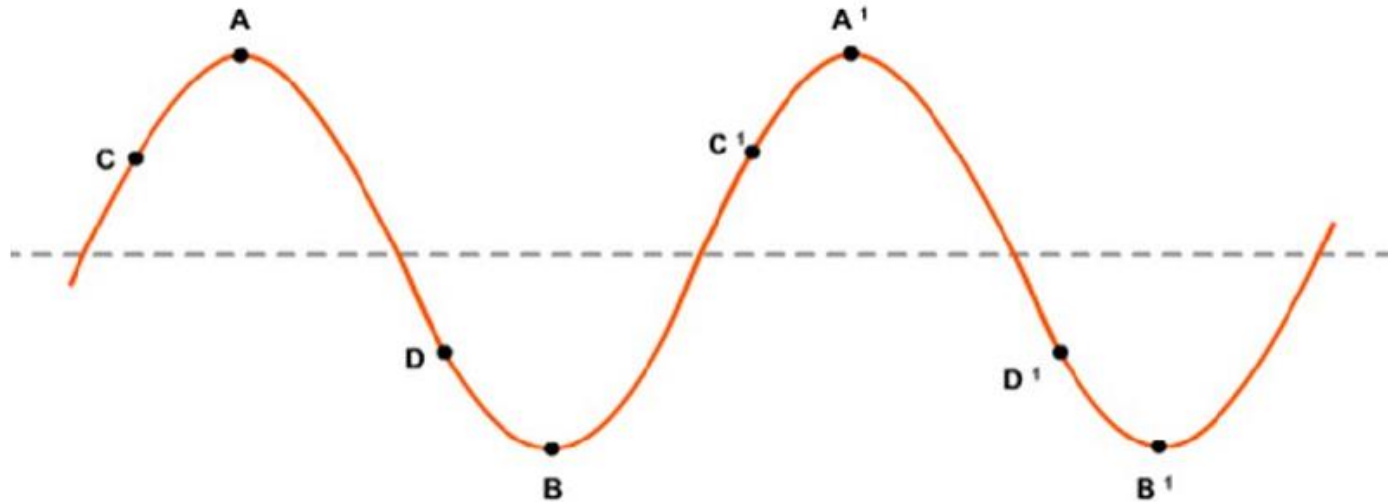
Wavelength,
Amplitude,
Frequency,
Phase



waves of various frequencies
the lower waves have higher frequencies than those above.

Phase of a wave

The phase of a wave relates the position of a feature, typically a peak or a trough of the waveform, to that same feature in another part of the waveform.



- ❑ The entire argument of the sine wave is known as the phase of the wave.
- ❑ Phase relates the spatial position of wave with time coordinate.

$$\psi = A \sin \frac{2\pi}{\lambda} (x \mp vt)$$

$$\psi = A \sin(kx - \omega t + \phi_0)$$

where ϕ_0 is known as initial phase

Phase difference & path difference

If the phase difference between two waves is 2π then the path difference between that two waves is λ .

Let for a path difference x , the phase difference is ϕ .

We know that for a path difference λ , phase difference = 2π

So for path difference x the phase difference would be:

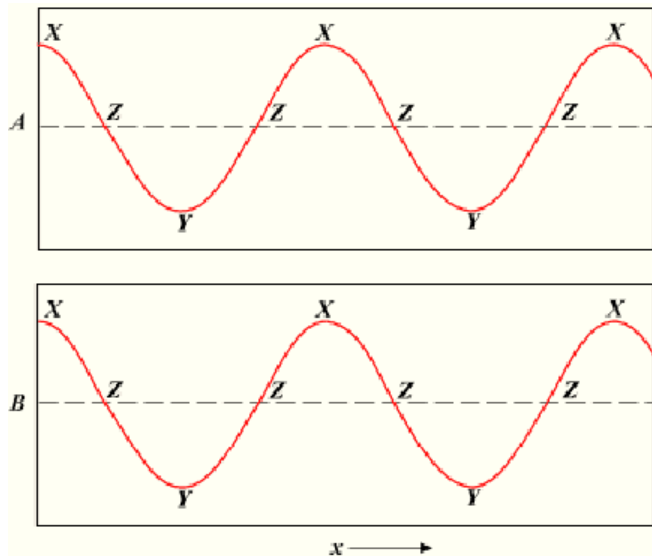
$$\phi = \frac{2\pi x}{\lambda}$$

$$\Rightarrow \text{Phase difference} = \frac{2\pi}{\lambda} \times \text{Path difference}$$

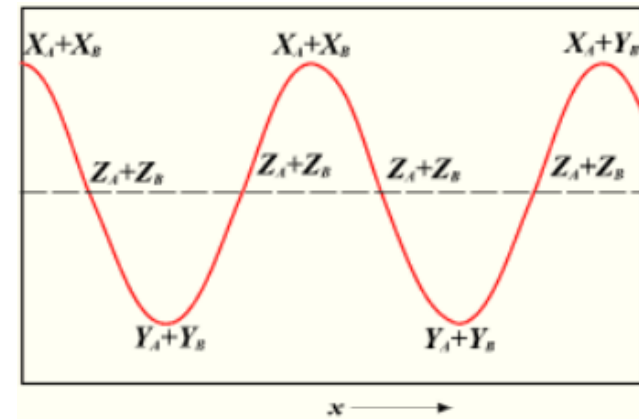
Phase difference

- ❑ The **phase difference** between two **signals** of the **same frequency** can be thought of as a delay or advance in the start of one signal's cycle with respect to another.
- ❑ Usually, it is expressed in the unit of angle.

Waves in-phase



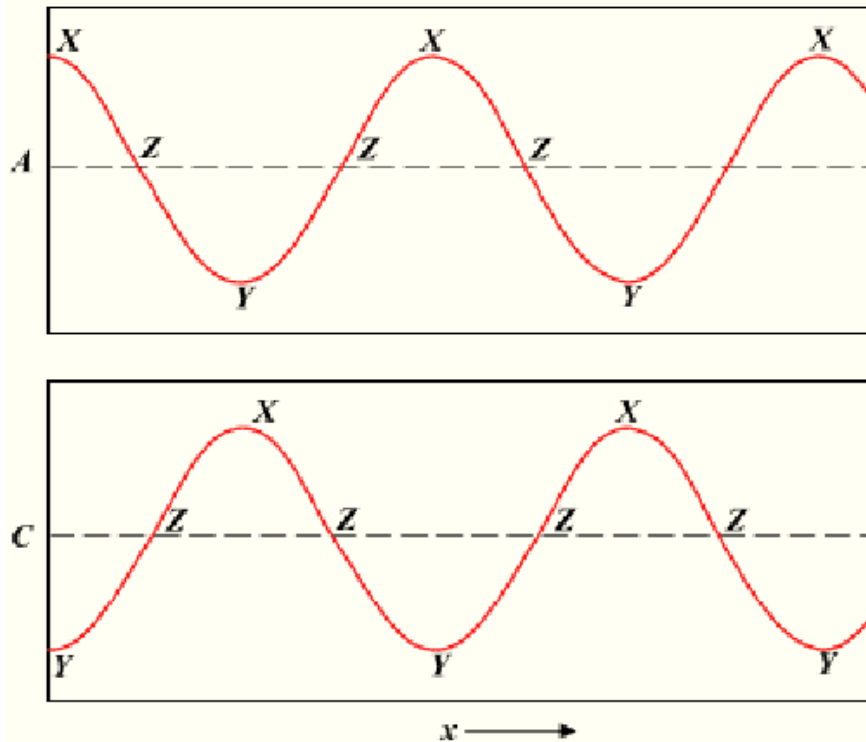
Combined waves



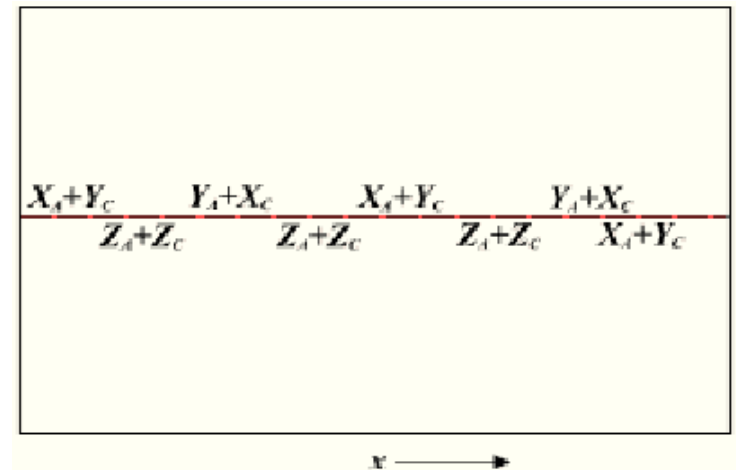
Positions of the peaks (X), troughs (Y) and zero-crossing points (Z) coincide. Thus **phase difference = 0** (Waves are in-phase).

Phase difference

Waves out -phase



Combined waves



Positions of the peaks (X), troughs (Y) and zero-crossing points (Z) are opposite. Thus **phase difference = 180° (Waves are out-phase).**

Wavefront

- It is the surface made up of particle having same phase during propagation of energy by method of oscillation or during propagation of wave.
- Direction of propagation of energy i.e. ray is always perpendicular to wavefront.
- The shape of wavefront depends on the shape of the source of disturbance.
- When the source of light is a point source the wavefront formed will be spherical wavefront, similarly cylindrical wavefront for line source and plane wavefront for the source at infinity.

Principle of Superposition

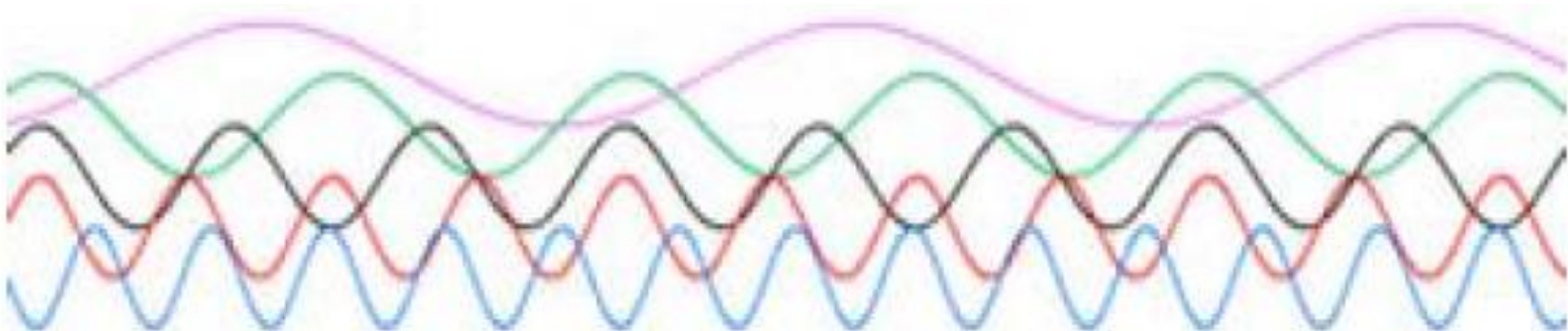
- When two separate waves arrive at the same place in space, they overlap and simply add (subtract) without permanently destroying or disrupting either wave.
- The resulting disturbance at each point in the region of overlap is the algebraic sum of the individual constituent waves at that location.

When two or more waves travel through the same medium simultaneously, the resultant displacement at a point is the vector sum of the displacement due to the individual waves.

Coherent Sources



*Sources, which emit waves (mostly light) of the same frequency, nearly same amplitude and also are always in phase with each other or having a constant phase relationship between them, known as **Coherent sources**.*



Multiple Choice Questions:

1) Nature of light is:[CO1]

- a) Longitudinal
- b) Transverse
- c) Longitudinal and Transverse
- d) None of these

2) Wave Nature of Light can be explained by :[CO1]

- a) Photoelectric effect
- b) Compton Effect
- c) Interference
- d) Diffraction

3) Coherent sources have [CO1]

- a) same frequency, same amplitude and constant Phase difference
- b) same frequency, different amplitude and constant Phase difference
- c) same Frequency, same amplitude (phase doesn't matter)
- d) None of the above

Multiple Choice Questions:

4) What is the relation between phase difference (ϕ) and path difference (x)? [CO2]

- a) $\phi = (2\pi/\lambda) * x$
- b) $\phi = (2\pi) * x$
- c) $\phi = (1/\lambda) * x$
- d) $\phi = (2/\lambda) * x$

5) If the source is in the form of point and line the wave front will be respectively, [CO1]

- a) Spherical and Cylindrical
- b) Concave and Convex
- c) Convex and Concave
- d) None of the above

Subjective Questions:

- What do you mean by wave and wavefront? [CO1]
- Discuss the four physical quantity used to describe a wave? [CO2]
- With the help of representative wave diagram explain the situation when two waves are in phase and out of phase? [CO2]
- Explain briefly the principle of superposition of waves? [CO2]
- What is coherent source? Can two independent sources be coherent? [CO2]